

Practical 1 - Introduction to Survival Analysis

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `pbcbase`, `surv_data_practical1`, `whitehall`

R packages required: `survival`, `mexhaz`, `flexsurv`

```
library(survival)
library(mexhaz)
library(flexsurv)
```

Introduction

This practical is in three parts.

Part A. Looks at some fundamentals of working with survival data. Using `pbcbase`.

Part B. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `surv_data_practical1`.

Part C. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `whitehall`.

Aims

The aims of this session are:

- Understand how survival data are stored and set-up for analysis.
- Derive the formula for the maximum likelihood estimator of the hazard rate parameter λ from an exponential distribution.

- Be able to obtain the maximum likelihood estimate $\hat{\lambda}$ from data.

Part A: Using survival data

The first dataset we will use is called `pbcbase`. This is data from a multicentre, double-blind, clinical trial for the treatment of Primary Biliary Cirrhosis (PBC). There are 184 individuals in the data and 17 variables; the variables of interest in this session are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>datein</code>	Date person entered the study
<code>dateout</code>	Date of the end of follow-up due to either death or censoring
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years

Question 1

Open the `pbcbase` dataset and familiarise yourself with the key variables.

SOLUTION

```
pbcbase<-read.table("datasets/pbcbase.csv",sep=";",header=T)
names(pbc)
```

```
[1] "id"      "datein"  "dateout" "time"    "d"       "treat"
[7] "age"     "logb0c"  "alb0c"   "logigm0c" "cenc0"   "cir0"
[13] "gh0"     "asc0"    "bil0"    "alb0"    "igm0"
```

Question 2

The `datein` and `dateout` variables are stored as text in the csv file. Start by formatting these as dates in R as follows: `pbcbase$datein=as.Date(pbc$datein,"%d%b%Y")` `pbcbase$dateout=as.Date(pbc$dateout,"%d%b%Y")` Take a look at what has happened to `datein` and `dateout`. They are now stored in the default form: YYYY-MM-DD. Behind the scenes in R the dates are now recorded as numbers. The number used by R to encode a particular date is as the number of days after 1st January 1970. So 1st January 1970 takes value 0, 2nd January 1970 takes value 1, etc.

The `lubridate` package is also useful for handling dates, but we do not use it in this module. Here is a helpful blog on dates and times in R: <https://www.gormananalysis.com/blog/dates-and-times-in-r-without-losing-your-sanity/>

SOLUTION

```
pbcc$datein=as.Date(pbcc$datein,"%d%b%Y")
pbcc$dateout=as.Date(pbcc$dateout,"%d%b%Y")

#note that R stores dates as number of days since 1 January 1970
as.numeric(as.Date("1970-01-02"))
```

```
[1] 1
```

Question 3

- How many people died in the dataset, and how many were censored?
- What was the earliest date of entry to the study? And the latest?
- What was the earliest date of exit from the study? And the latest?
- What happened to the first two people in the data (id=5 and id=23).

SOLUTION

```
table(pbcc$d)
```

```
0  1
88 96
```

```
range(pbcc$datein)
```

```
[1] "1980-01-03" "1980-12-25"
```

```
range(pbcc$dateout)
```

```
[1] "1980-01-19" "1992-06-26"
```

```
pbcc[1:2,c("id","datein","dateout","time","d")]
```

```

  id      datein      dateout      time d
1  5 1980-02-19 1981-09-19 1.582478 1
2 23 1980-07-22 1989-08-02 9.032169 0

```

- d is the event indicator, taking value 1 for individuals who die and 0 for individuals who are censored at their end of follow-up. Hence there were 96 deaths during the study follow-up, and 88 people were censored.
- The earliest date of entry was 3rd January 1980, and the latest was 25th December 1980.
- Exit dates ranged from 19th January 1980 to 26th June 1992.
- The person with id=5 joined the study on 19/2/1980 and died on 19/9/1981 (because d=1), which was a follow-up time of 1.58 years.
- The person with id=23 joined the study on 22/7/1980 and their censoring date (because d=0) was on 02/08/1989, which was a follow-up time of 9.03 years. That is, this person was known to be still alive after 9.03 years of follow-up.

Question 4

The fact that R stores dates as numbers makes calculating the time between two dates simple. To calculate the length of time each person was followed in the study (which we call `days_in_study`) we can type: `pbcs$days_in_study = as.numeric(pbc$dateout - pbc$datein)`

Compare this new variable to the `time` variable to make sure this has done what you expect.

SOLUTION

```

#calculate difference between datein and dateout
pbcs$days_in_study = as.numeric(pbc$dateout - pbc$datein)
summary(pbc$days_in_study)

```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
8	611	1314	1483	2190	4253

```

#days_in_study was in days whereas time is in years
#calculate difference between datein and dateout in years
pbcs$years_in_study = as.numeric(pbc$dateout - pbc$datein)/365.25

#compare years_in_study and time
summary(pbc[,c("years_in_study", "time")])

```

years_in_study	time
Min. : 0.0219	Min. : 0.0219
1st Qu.: 1.6728	1st Qu.: 1.6728
Median : 3.5962	Median : 3.5962
Mean : 4.0598	Mean : 4.0600
3rd Qu.: 5.9945	3rd Qu.: 5.9945
Max. : 11.6441	Max. : 11.6441

- The `time` variable records the follow-up time (from `datein` to `dateout`) in years.
- When obtaining the difference between two dates we get that in days. This can be converted to years by dividing by 365.25 (this is a common way of handling leap years).

Question 5

What do you think is the appropriate time origin in the PBC study?

In R the `survival` package contains the key tools for analysis of survival data. The `Surv()` function is fundamental to survival analysis in R. It is used in functions such as model formulas to specify the time periods over which people are observed and to specify which individuals have the event and which are censored. Try the following: `Surv(time=pbcs$time,event=pbcs$d)`

Take a look at the output. What do the `+` symbols represent? This method uses the date of randomisation as the time origin (i.e. time since randomisation is the time scale) and specifies the event indicator. As the `time` variable is in years, this will be the units in which time is used in any analysis in which the the above `Surv()` function is included.

Another way of specifying the `Surv()` function is by using the dates:

```
Surv(time=as.numeric(pbc$dateout),event=pbc$d,origin=as.numeric(pbc$datein))
```

In this version, the time of the end of follow-up is provided as the date each person left the study, and the options tell R the time origin (the date of joining the study, in this case), and the event indicator. Because the entry and exit dates (`datein` and `dateout`) are stored in number of days, the unit of analysis will be days. Note that the `Surv()` function does not accept times that are formatted as dates, which is why we use `as.numeric` above.

SOLUTION

```
#Just showing the first 10 values in the interests of space
# Surv(time=pbcs$time,event=pbcs$d)
Surv(time=pbcs$time,event=pbcs$d)[1:10,]
```

```
[1] 1.582478 9.032169+ 2.286105 2.078029 5.311431 2.633812+ 3.931554+
[8] 4.843258+ 2.105407+ 2.819986+
```

```
#Just showing the first 10 values in the interests of space
# Surv(time=as.numeric(pbc$dateout),event=pbc$d,origin=as.numeric(pbc$datein))
Surv(time=as.numeric(pbc$dateout),event=pbc$d,
      origin=as.numeric(pbc$datein))[1:10,]
```

```
[1] 578 3298+ 835 759 1940 962+ 1436+ 1769+ 769+ 1030+
```

- As this is a clinical trial, the date of randomisation is the most appropriate time origin.
- The plus signs in the output from `Surv()` indicate censoring times.
- We don't usually look at the output from `Surv()` like this as part of an analysis, but it is useful here to see how it is recording the time-to-event data.

Part B: Fitting models to survival data

Question 1

Consider a study of n individuals. Let t_i denote the event or censoring time for individual i and let δ_i be the indicator of whether the person died (meaning t_i is their time of death) or was censored (meaning t_i is their censoring time, i.e. they were known to be alive at this time).

Write down the likelihood for survival data (including censoring) assumed to follow an exponential distribution with hazard λ . Derive a formula for the maximum likelihood estimate for λ .

SOLUTION

The likelihood is

$$L = \prod_{i=1}^n h(t_i)^{\delta_i} S(t_i) = \prod_{i=1}^n \lambda^{\delta_i} \exp(-\lambda t_i),$$

where $h(t)$ denotes the hazard at time t and $S(t)$ denotes the survival function at time t .

The log likelihood is therefore

$$l = \sum_{i=1}^n \delta_i \log \lambda - \lambda \sum_{i=1}^n t_i.$$

Differentiating the log likelihood with respect to λ gives

$$\frac{dl}{d\lambda} = \sum_{i=1}^n \delta_i / \lambda - \sum_{i=1}^n t_i.$$

The maximum likelihood estimator for λ is therefore $\hat{\lambda} = \sum_{i=1}^n \delta_i / \sum_{i=1}^n t_i$, which is the total number of events (deaths) divided by the total person-time of follow-up.

Question 2

We will now use the formula derived for $\hat{\lambda}$ in a simulated dataset called `surv_data_practical1`. There are just two variables: `survtimes` contains the survival time for each participant, and `d` the outcome. In this dataset all participants experienced the event (`d=1`). The data were simulated for 100 individuals, with event times generated from an exponential distribution.

Load the data using

```
dta=read.table("datasets/surv_data_practical1.csv",sep="," ,header=T)
```

Using your formula from Part B, Question 1 calculate the maximum likelihood estimate for λ and interpret the estimate.

SOLUTION

```
sum(dta$d)
```

```
[1] 100
```

```
sum(dta$survtimes)
```

```
[1] 460.2566
```

```
lambda.est<-sum(dta$d)/sum(dta$survtimes)  
lambda.est
```

```
[1] 0.2172701
```

In these data, $\sum_{i=1}^n \delta_i = 100$ and $\sum_{i=1}^n t_i = 460.2566$. Thus the estimate of λ is $\hat{\lambda} = 0.217$. The interpretation is that the estimated hazard of mortality is 0.217 per year of follow-up. If we observed 1000 people for 1 year we would expect to see 217 deaths. If we observed 2000 people for 6 months we would expect to see 217 deaths. If we followed 500 individuals for 2 years we would expect to see 217 deaths. Because we used an exponential model, the hazard is assumed to be the same at all times, i.e. it does not change over time.

Question 3

We will fit an exponential model to these data to confirm the maximum likelihood estimate for λ that we calculated above. Compare the results with what you found above. In R exponential models can be fitted using the `survreg` function from the `survival` package:

```
exp.model = survreg(Surv(survtimes)~1,dist="exponential",data=mydata)
summary(exp.model)
```

It is important to note that the `survreg` function with `dist="exponential"` estimates $-\log \lambda$ rather than λ (See Table 3.1 in the lecture notes).

Another option is the `mexhaz` package, where we can fit an exponential distribution using:

```
exp.model2 = mexhaz(Surv(survtimes)~1,base="pw.cst",data=dta)
exp.model2
```

The `flexsurv` package provides another way of fitting an exponential model, as follows:

```
exp.model3 = flexsurvreg(Surv(survtimes)~1,dist="exponential",data=mydata)
exp.model3
```

Compare the output given by the different functions.

SOLUTION

```
#using survreg
exp.model = survreg(Surv(survtimes)~1,dist="exponential",data=dta)
summary(exp.model)
```

Call:

```
survreg(formula = Surv(survtimes) ~ 1, data = dta, dist = "exponential")
```

	Value	Std. Error	z	p
(Intercept)	1.53	0.10	15.3	<2e-16

Scale fixed at 1

Exponential distribution

Loglik(model)= -252.7 Loglik(intercept only)= -252.7

Number of Newton-Raphson Iterations: 5

n= 100

```
# using mexhaz
exp.model2 = mexhaz(Surv(survtimes)~1,base="pw.cst",data=dta,
                    print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
dta	100	100	100	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
5	16	pw.cst	20	10	nlm	---	1	-252.6614	0.02

```
exp.model2
```

Call:

```
mexhaz(formula = Surv(survtimes) ~ 1, data = dta, base = "pw.cst",
        print.level = 0)
```

Coefficients:

	Estimate	StdErr	z.value	p.value
[0,20.4]	-1.5266	0.1000	-15.266	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

log-likelihood: -252.6614 (for 1 degree(s) of freedom)

number of observations: 100, number of events: 100

```
#using flexsurvreg
exp.model3 = flexsurvreg(Surv(survtimes)~1,dist="exponential",data=dta)
exp.model3
```

Call:

```
flexsurvreg(formula = Surv(survtimes) ~ 1, data = dta, dist = "exponential")
```

Estimates:

	est	L95%	U95%	se
rate	0.2173	0.1786	0.2643	0.0217

N = 100, Events: 100, Censored: 0
Total time at risk: 460.2566
Log-likelihood = -252.6614, df = 1
AIC = 507.3228

- The estimate of λ is 0.217, which is what we found in our by-hand calculation above.
- The `survreg` output gives an estimate of $-\log \lambda$, which is 1.53.
- The `mexhaz` output gives an estimate of $\log \lambda$, which is -1.53.
- The `flexsurvreg` output gives an estimate of λ , which is 0.217.
- See Table 3.1 in the lecture notes.

Part C: Whitehall study data

The Whitehall Study is a cohort study, which was started in 1967 and is a cohort of male civil servants employed in various government departments in London. It has been used to study risk factors for mortality in men. We will focus on deaths due to coronary heart disease (CHD). The dataset contains 1677 individuals and 14 variables, with the variables used in this practical summarised below:

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>timebth</code>	Date of birth
<code>timein</code>	Date person entered the study
<code>timeout</code>	Date of the end of follow-up due to either death or censoring
<code>chd</code>	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease

Question 1

- Load the data and format `timebth`, `timein` and `timeout` as dates.
- What is the latest date of the end the follow-up? How many people had their end of follow-up on this date? What could the explanation for this be?

SOLUTION

```
#load data
whl=read.table("datasets/whitehall.csv",sep="," ,header=T)

#format dates
```

```

whl$timein=as.Date(whl$timein,"%d%b%Y")
whl$timeout=as.Date(whl$timeout,"%d%b%Y")
whl$timebth=as.Date(whl$timebth,"%d%b%Y")

```

```

#latest date of exit from the study?

```

```

range(whl$timeout)

```

```

[1] "1968-11-27" "1987-01-30"

```

```

#Number of with end of follow-up on 1987-01-30

```

```

sum(whl$timeout=="1987-01-30")

```

```

[1] 1275

```

- The latest date of the end the follow-up is 30th January 1987, when 1275 people are listed as being censored. This is most likely the date on which the study closed and follow-up was stopped, or the date this dataset was extracted.

Question 2

This is an example of data where event times could be left-truncated, meaning that not everyone is observed from the time origin of interest. In this type of study we could be interested in the date of joining the study as the time origin, or the date of birth as the time origin, depending on our question.

Compare the following four ways of using `Surv()` for this data.

```

#(i) time origin: date of joining cohort
#time scale: time in study
Surv(time=as.numeric(whl$timeout),event=whl$chd,origin=as.numeric(whl$timein))

#(ii) time origin: date of birth
#time scale: age
#delayed entry: ignored (not a good idea)
Surv(time=as.numeric(whl$timeout),event=whl$chd,origin=as.numeric(whl$timebth))

#(iii) time origin: date of birth
#time scale: age
#delayed entry: left truncation accommodated by providing the
# information that people join the study at timein
Surv(time=as.numeric(whl$timein),

```

```
time2=as.numeric(whl$timeout),  
event=whl$chd,  
origin=as.numeric(whl$timebth))
```

SOLUTION

- Approach (i) uses time-in-study as the timescale. Approaches (ii) and (iii) use age as the time scale, but (ii) does not account for the delayed entry/left truncation, which is incorrect. In (ii) individuals are being considered to be ‘at risk’ before they joined the cohort, which we should not allow. If there is delayed entry we should account for that in the analysis.